

Experiment (11)

Virtual Local Area Network (VLAN)

11.1 Objectives

To achieve Importance Of VLAN

11.2 Equipment / Program Requirements:

eNSP

Cisco Packet Tracker

11.3 Experiment Setup:

Draw your topo

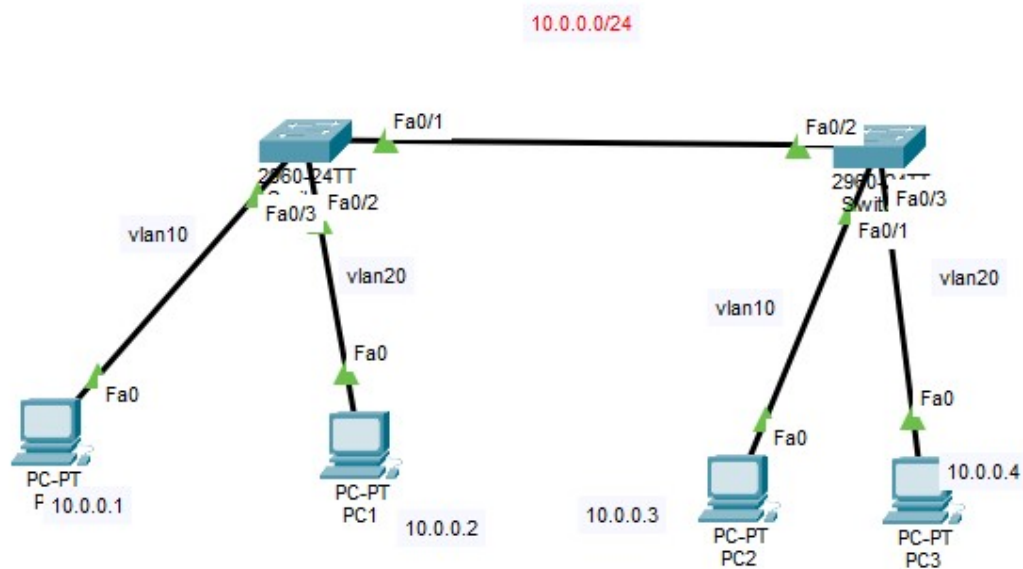
Then make configuration

Then display vlans in topology

11.4 Simulation Results:

Cisco Packet Tracker:

Topology:



SW 0:

```

Switch(config)#
Switch(config)#int fa0/3
Switch(config-if)#s
Switch(config-if)#swi
Switch(config-if)#switchport mode ac
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switch mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#do sh
Switch(config)#do sh vlan

```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10 VLAN0010	active	Fa0/3
20 VLAN0020	active	Fa0/2
1002 fddi-default	active	

Switch0

Physical Config CLI Attributes

IOS Command Line Inter

```
Switch(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#do sh
Switch(config)#do sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/3
20	VLAN0020	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0

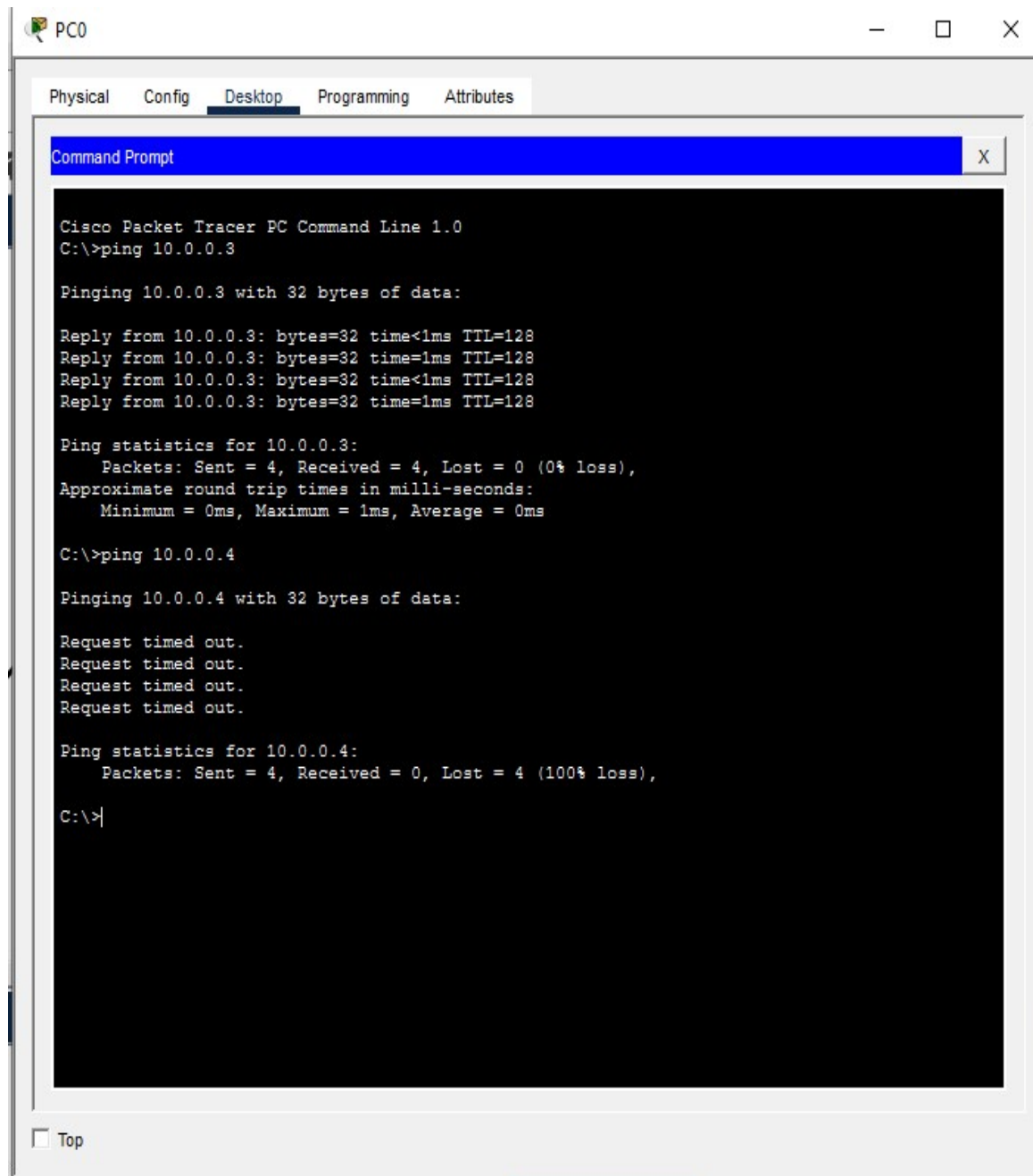
```
Switch(config)#
```

SW1:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#sw
Switch(config-if)#switchport mode access
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if)#exit
Switch(config)#sw
Switch(config)#int fa0/3
Switch(config-if)#sw
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#do sh
Switch(config)#do sh vlan
```

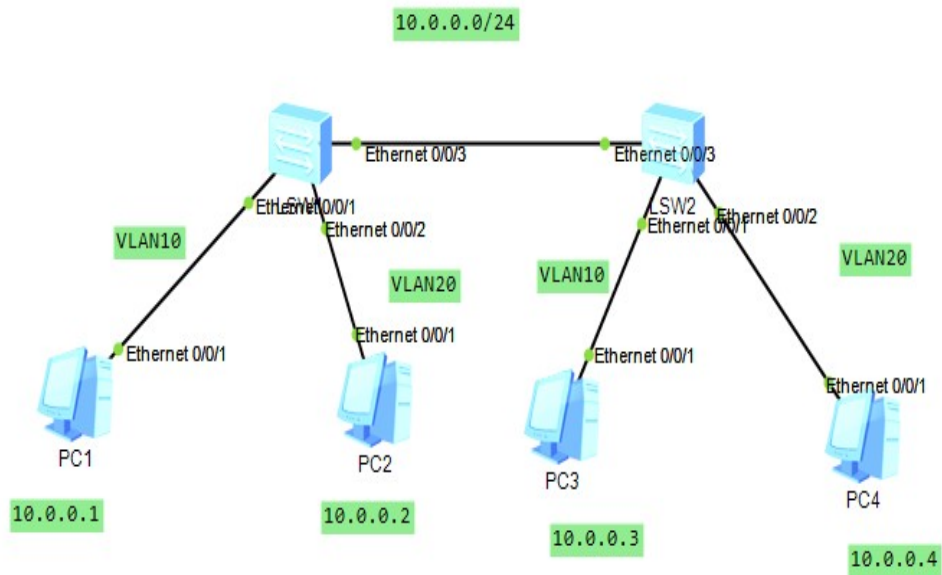
VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/1
20	VLAN0020	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Testing:



eNSP:

Topology:



SW1:

The device is running!

<Huawei>u t l

Info: Current terminal logging is off.

<Huawei>u t m

Info: Current terminal monitor is off.

<Huawei>int eth0/0/1

^

Error: Unrecognized command found at '^' position.

<Huawei>sys

Enter system view, return user view with Ctrl+Z.

[Huawei]int eth0/0/1

[Huawei-Ethernet0/0/1]q

[Huawei]vlan batc

[Huawei]vlan batch 10 20

Info: This operation may take a few seconds. Please wait for a moment...done.

[Huawei]int eth0/0/1

[Huawei-Ethernet0/0/1]por

[Huawei-Ethernet0/0/1]port 1

[Huawei-Ethernet0/0/1]port link-t

[Huawei-Ethernet0/0/1]port link-type access

[Huawei-Ethernet0/0/1]port def

[Huawei-Ethernet0/0/1]port default vlan 10

[Huawei-Ethernet0/0/1]q

[Huawei]int eth0/0/2

[Huawei-Ethernet0/0/2]port lin

[Huawei-Ethernet0/0/2]port link-ty

[Huawei-Ethernet0/0/2]port link-type access

[Huawei-Ethernet0/0/2]port default vlan20

^

Error: Unrecognized command found at '^' position.

[Huawei-Ethernet0/0/2]port default vlan 20

[Huawei-Ethernet0/0/2]q

[Huawei]int eth0/0/3

[Huawei-Ethernet0/0/3]port li

[Huawei-Ethernet0/0/3]port link-ty

[Huawei-Ethernet0/0/3]port link-type trunk

[Huawei-Ethernet0/0/3]port trunk allow-

[Huawei-Ethernet0/0/3]port trunk allow-pass vlan 10 20

[Huawei-Ethernet0/0/3]q

[Huawei]

Error: Ambiguous command found at '^' position.

```
[Huawei-Ethernet0/0/2]
[Huawei-Ethernet0/0/2]port link defa
[Huawei-Ethernet0/0/2]port def
[Huawei-Ethernet0/0/2]port default vlan 20
[Huawei-Ethernet0/0/2]q
[Huawei]int eth0/0/3
[Huawei-Ethernet0/0/3]port lin
[Huawei-Ethernet0/0/3]port link-ty
[Huawei-Ethernet0/0/3]port link-type trunk
[Huawei-Ethernet0/0/3]port trunk all
[Huawei-Ethernet0/0/3]port trunk allow-pass vlan 10 20
[Huawei-Ethernet0/0/3]q
[Huawei]display vlan
The total number of vlans is : 3
```

```
-----
U: Up;           D: Down;           TG: Tagged;      UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
-----
```

```
VID  Type  Ports
-----
```

```
1    common  UT:Eth0/0/3(U)   Eth0/0/4(D)   Eth0/0/5(D)   Eth0/0/6(D)
                Eth0/0/7(D)   Eth0/0/8(D)   Eth0/0/9(D)   Eth0/0/10(D)
                Eth0/0/11(D)  Eth0/0/12(D)  Eth0/0/13(D)  Eth0/0/14(D)
                Eth0/0/15(D)  Eth0/0/16(D)  Eth0/0/17(D)  Eth0/0/18(D)
                Eth0/0/19(D)  Eth0/0/20(D)  Eth0/0/21(D)  Eth0/0/22(D)
                GE0/0/1(D)   GE0/0/2(D)
```

```
10   common  UT:Eth0/0/1(U)
                TG:Eth0/0/3(U)
20   common  UT:Eth0/0/2(U)
                TG:Eth0/0/3(U)
```

```
VID  Status  Property  MAC-LRN  Statistics  Description
-----
```

```
1    enable  default   enable   disable    VLAN 0001
10   enable  default   enable   disable    VLAN 0010
20   enable  default   enable   disable    VLAN 0020
```

```
[Huawei]
```

SW2:

```
[Huawei]int eth0/0/1
[Huawei-Ethernet0/0/1]port lin
[Huawei-Ethernet0/0/1]port link-ty
[Huawei-Ethernet0/0/1]port link-type access
[Huawei-Ethernet0/0/1]port default vlan 10
Error: The VLAN does not exist.
[Huawei-Ethernet0/0/1]6
      ^
Error: Unrecognized command found at '^' position.
[Huawei-Ethernet0/0/1]q
[Huawei]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[Huawei]int eth0/0/1
[Huawei-Ethernet0/0/1]port lin
[Huawei-Ethernet0/0/1]port link-typo
[Huawei-Ethernet0/0/1]port link-typ
[Huawei-Ethernet0/0/1]port link-type access\
      ^
Error: Unrecognized command found at '^' position.
[Huawei-Ethernet0/0/1]port link-type access
[Huawei-Ethernet0/0/1]port def
[Huawei-Ethernet0/0/1]port default vlan 10
[Huawei-Ethernet0/0/1]q
[Huawei]int eth0/0/2
[Huawei-Ethernet0/0/2]port link
[Huawei-Ethernet0/0/2]port link-ty
[Huawei-Ethernet0/0/2]port link-type access
[Huawei-Ethernet0/0/2]port default vlan 20
[Huawei-Ethernet0/0/2]q
[Huawei]int eth0/0/3
[Huawei-Ethernet0/0/3]port lin
[Huawei-Ethernet0/0/3]port link-typ
[Huawei-Ethernet0/0/3]port link-type tru
[Huawei-Ethernet0/0/3]port link-type trunk
[Huawei-Ethernet0/0/3]port trunk allow
[Huawei-Ethernet0/0/3]port trunk allow-pass vlan 10 20
[Huawei-Ethernet0/0/3]q
[Huawei]display vlan
The total number of vlans is : 3
-----
U: Up;          D: Down;          TG: Tagged;      UT: Untagged;
MP: Vlan-mapping;  ST: Vlan-stacking;
#: ProtocolTransparent-vlan;  *: Management-vlan;
-----
```

```
[Huawei]int eth0/0/2
[Huawei-Ethernet0/0/2]port link
[Huawei-Ethernet0/0/2]port link-ty
[Huawei-Ethernet0/0/2]port link-type access
[Huawei-Ethernet0/0/2]port default vlan 20
[Huawei-Ethernet0/0/2]q
[Huawei]int eth0/0/3
[Huawei-Ethernet0/0/3]port lin
[Huawei-Ethernet0/0/3]port link-typ
[Huawei-Ethernet0/0/3]port link-type tru
[Huawei-Ethernet0/0/3]port link-type trunk
[Huawei-Ethernet0/0/3]port trunk allow
[Huawei-Ethernet0/0/3]port trunk allow-pass vlan 10 20
[Huawei-Ethernet0/0/3]q
[Huawei]display vlan
The total number of vlans is : 3
```

```
-----
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;
MP: Vlan-mapping;   ST: Vlan-stacking;
#: ProtocolTransparent-vlan;  *: Management-vlan;
-----
```

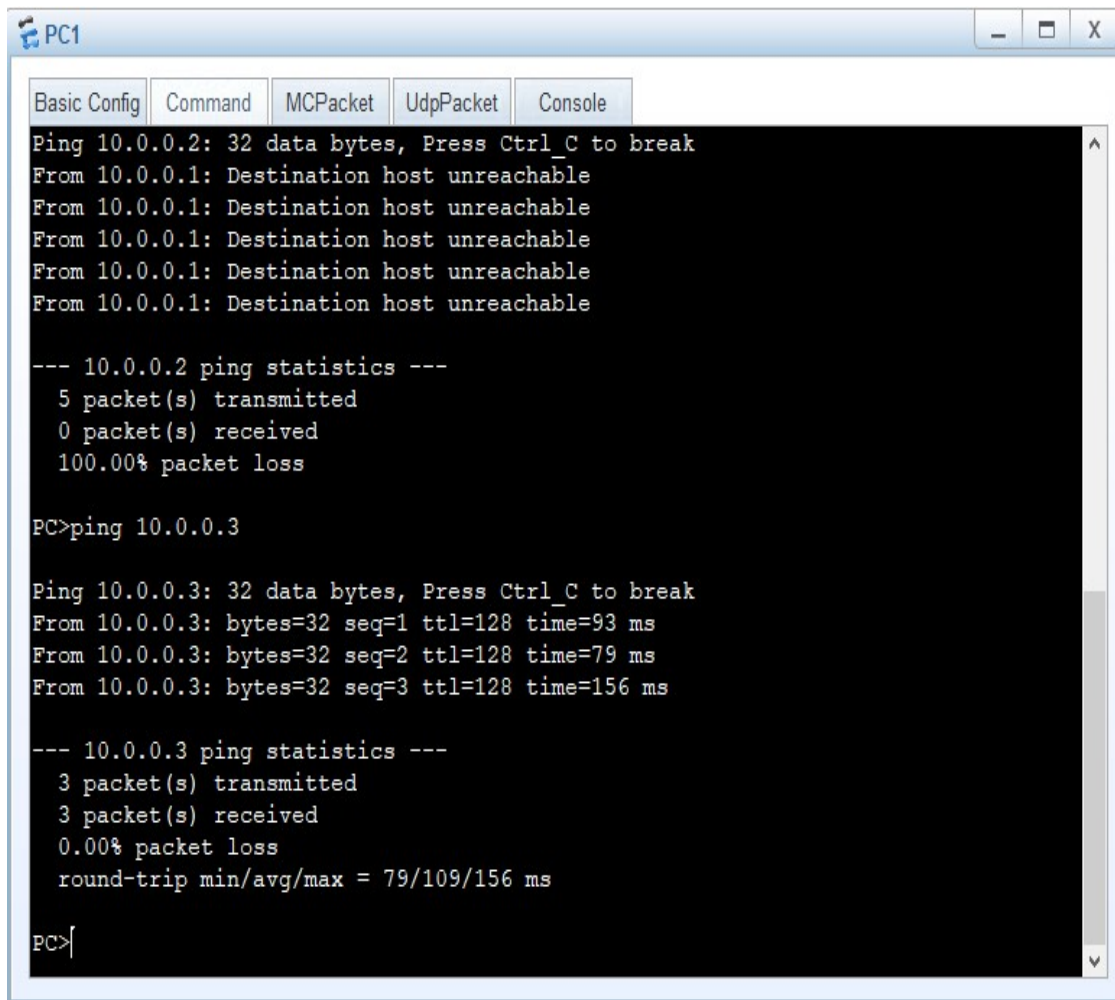
```
-----
VID  Type  Ports
-----
1    common  UT:Eth0/0/3(U)   Eth0/0/4(D)   Eth0/0/5(D)   Eth0/0/6(D)
                Eth0/0/7(D)   Eth0/0/8(D)   Eth0/0/9(D)   Eth0/0/10(D)
                Eth0/0/11(D)  Eth0/0/12(D)  Eth0/0/13(D)  Eth0/0/14(D)
                Eth0/0/15(D)  Eth0/0/16(D)  Eth0/0/17(D)  Eth0/0/18(D)
                Eth0/0/19(D)  Eth0/0/20(D)  Eth0/0/21(D)  Eth0/0/22(D)
                GE0/0/1(D)   GE0/0/2(D)

10   common  UT:Eth0/0/1(U)
                TG:Eth0/0/3(U)

20   common  UT:Eth0/0/2(U)
                TG:Eth0/0/3(U)
```

```
-----
VID  Status  Property  MAC-LRN  Statistics  Description
-----
1    enable  default  enable  disable  VLAN 0001
10   enable  default  enable  disable  VLAN 0010
20   enable  default  enable  disable  VLAN 0020
[Huawei]
```

Testing:



The screenshot shows a PC1 console window with a title bar and standard window controls. It has five tabs: 'Basic Config', 'Command', 'MCPacket', 'UdpPacket', and 'Console'. The 'Console' tab is active, displaying a series of ping commands and their results. The first command is 'ping 10.0.0.2', which results in five 'Destination host unreachable' messages. This is followed by 'ping statistics' for 10.0.0.2, showing 5 packets transmitted, 0 received, and 100% packet loss. The second command is 'ping 10.0.0.3', which results in three successful ping responses with varying times (93, 79, and 156 ms). This is followed by 'ping statistics' for 10.0.0.3, showing 3 packets transmitted and received, 0% packet loss, and a round-trip time of 79/109/156 ms. The console ends with a 'PC>' prompt.

```
PC1
Basic Config Command MCPacket UdpPacket Console
Ping 10.0.0.2: 32 data bytes, Press Ctrl_C to break
From 10.0.0.1: Destination host unreachable
From 10.0.0.1: Destination host unreachable
From 10.0.0.1: Destination host unreachable
From 10.0.0.1: Destination host unreachable
From 10.0.0.1: Destination host unreachable

--- 10.0.0.2 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss

PC>ping 10.0.0.3

Ping 10.0.0.3: 32 data bytes, Press Ctrl_C to break
From 10.0.0.3: bytes=32 seq=1 ttl=128 time=93 ms
From 10.0.0.3: bytes=32 seq=2 ttl=128 time=79 ms
From 10.0.0.3: bytes=32 seq=3 ttl=128 time=156 ms

--- 10.0.0.3 ping statistics ---
 3 packet(s) transmitted
 3 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 79/109/156 ms

PC>|
```

11.5 Conclusions:

Virtual Local Area Networks (VLANs) play a crucial role in modern network design by logically segmenting a physical network into smaller, isolated broadcast domains. This segmentation enhances network performance by reducing unnecessary traffic and improves security by restricting access between different groups of devices. VLANs also provide flexibility, allowing network administrators to organize devices based on function, department, or policy rather than physical location. Overall, VLANs contribute to efficient network management, scalability, and better control over data flow, making them an essential component in enterprise networking.